

Overfitting vs. Underfitting

- **Definitions:**

Overfitting: Occurs when a machine learning model learns the training data *too well*, memorizing its noise, outliers, and specific details rather than discovering general patterns. As a result, the model achieves extremely high performance (low error) on training data but fails to generalize, performing poorly (high error) on unseen test data.

Underfitting: Occurs when a machine learning model is too simplistic to capture the underlying structure or relationships within the data. The model performs poorly on both the training data and unseen test data, indicating that it has failed to learn the domain logic properly.

- **Main Causes:**

Causes of Overfitting

- **Excessive Model Complexity:** Using a model with too many parameters (e.g., high-degree polynomials, overly deep decision trees) relative to the task.
- **Insufficient Training Data:** Training a complex model on a small dataset, making it easy for the model to memorize instances.
- **Noisy Data:** High amounts of uncleaned data or outliers that the model attempts to fit precisely.
- **Over-training:** Training an iterative model (e.g., neural network) for too many epochs without early stopping.

Causes of Underfitting

- **Model Oversimplification:** Choosing a model that lacks capacity (e.g., using a linear model for a strongly non-linear dataset).
- **Insufficient Features:** Providing too few input features to represent the problem domain adequately.
- **Over-regularization:** Applying excessive regularization constraints (e.g., L1/L2 penalty parameters being too high), which restricts the model parameters excessively.
- **Premature Training Stop:** Stopping model training before it converges on an optimal cost function minimum.

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